

● MEDIUM

● INFORMATION AND IDEAS

● ANSWER KEY

Inferences

02

10 answers · Rationale-rich · Teach from doc

Q1**C**

C. the mothers of the two individuals probably reproduced through parthenogenesis.

Choice C is the best answer because it most logically completes the text's discussion of parthenogenesis as it relates to two California condors at the San Diego Zoo. The text explains that parthenogenesis is a form of reproduction in which a female reproduces without mating, producing an egg containing an offspring with DNA that comes entirely from its mother. After noting examples of parthenogenesis among birds, the text then reports on a discovery at the San Diego Zoo. According to the text, scientists at the zoo found that two of the zoo's California condors weren't similar enough to any of the zoo's male condors to be their offspring. Given that parthenogenesis occurs without mating and results in an offspring with DNA entirely from its mother, it can reasonably be inferred that the scientists concluded that the mothers of the two condors probably didn't mate with any of the male condors at the zoo, reproducing through parthenogenesis instead.

Choice A is incorrect because the text never suggests that at least one of the condors' mothers was born as a result of parthenogenesis. Rather, the text suggests that the two condors were themselves both probably born as a result of parthenogenesis, which would account for their lack of strong genetic resemblance to any of the male condors in the zoo's condor enclosure. Choice B is incorrect because the text never discusses wild California condors, only California condors living at the San Diego Zoo. For this reason, the text doesn't support a conclusion that distinguishes between how California condors reproduce in zoos and how they reproduce in the wild. Choice D is incorrect because the text never suggests that the zoo's female condors lacked sufficient access to males or that this would be a cause of parthenogenesis, indicating only that the condor enclosure included males.

Q2

B

B. has been preserved in certain mammal species because it benefits their immune systems.

Choice B is the best answer because it most logically completes the text's discussion of Smith and colleagues' investigation of the evolution and biological role of the appendix. The text indicates that the team found several instances of the appendix emerging and not disappearing in the lineages of various mammal species the team examined. Furthermore, the text states that species that possess an appendix also tend to have relatively high amounts of lymphoid tissue—a type of tissue that supports immune system function. Taken together, these details strongly support the hypothesis that the appendix has persisted in some species because it has a function that contributes to effective immune responses in those species.

Choice A is incorrect because the text doesn't address any nonmammalian species. Choice C is incorrect because the text doesn't make predictions about the evolutionary future of the species Smith and colleagues examined, and although the implication of the text is that the appendix likely does serve a function for the immune system, nothing in the text indicates that the appendix will become more widespread in the future. Choice D is incorrect. Although the text does suggest an association between having an appendix and relatively high concentrations of lymphoid tissue, it doesn't claim that the appendix causes the tissue to grow, nor does it address the relative production of the tissue at different periods of time.

Q3

A

- A. the current makeup of the Arabia Terra region might not accurately reflect the volcanic activity of Mars's past.
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Choice A is the best answer because it presents the conclusion that most logically follows from the text's discussion of the Arabia Terra region of Mars. According to the text, there are craters in Arabia Terra that could be the result of volcanic activity, and researchers have found evidence of ash deposits consistent with a large eruption. The text goes on to note, however, that erosion and other events could have altered the surface of Mars. This observation suggests that current conditions on Mars's surface are not necessarily a reliable guide to past events—some signs of past events could have been transformed or erased entirely—and thus the current makeup of Arabia Terra may not accurately reflect past volcanic activity.

Choice B is incorrect because the text suggests only that past events could have changed Mars's surface such that its current appearance isn't a reliable guide to past activity, not that it's likely that past eruptions were not as massive as astronomers previously believed. Nothing in the text supports a conclusion about the likely size of past eruptions. Choice C is incorrect because the observation that resurfacing events could have changed the appearance of Mars doesn't imply that the ash discussed in the text likely came from multiple volcanoes. Although it's possible that the ash came from different volcanoes, there's no information in the text supporting a conclusion about how likely that possibility is. Choice D is incorrect because nothing in the text suggests that the Arabia Terra craters had to have been created by something other than volcanic eruptions. Although the text does suggest that the evidence consistent with volcanic eruptions shouldn't be taken as definitive proof of past eruptions, that doesn't mean that the craters couldn't have been created by eruptions, only that we can't be certain they were.

Q4

D

D. reducing invasive *E. esula* numbers without harming other organisms.

Choice D is the best answer because it presents the conclusion that most logically follows from the text’s discussion of leafy spurge and engineered DNA. The text establishes that using chemical herbicides to control leafy spurge in North America can also harm other plants nearby. The text then indicates that it might be possible to use engineered DNA to prevent plants from reproducing, which would be useful for “exclusively targeting” leafy spurge. If it’s possible to exclusively target leafy spurge with engineered DNA—meaning that only leafy spurge is affected by the engineered DNA—and prevent the plant from reproducing, then leafy spurge numbers could be reduced “without harming other organisms.”

Choice A is incorrect because the text raises the possibility of using engineered DNA to prevent leafy spurge from reproducing, not to make individual leafy spurge plants more vulnerable to chemical herbicides that already exist. Choice B is incorrect because the text doesn’t describe any ecological benefits of leafy spurge in North America; instead, the text is focused on using engineered DNA to prevent leafy spurge from reproducing and thereby reduce its numbers. The only ecological effects of leafy spurge in North America that are described in the text are harmful. Choice C is incorrect because the text describes the possibility of using engineered DNA to prevent leafy spurge from reproducing; it doesn’t offer a way to enable cattle to eat leafy spurge without becoming sick.

Q5

B

- B. knowing an ancient civilization's language isn't necessary in order to learn details about the civilization.
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Choice B is the best answer because it most logically completes the text's discussion of the Indus River valley civilization. The text establishes that archaeologists haven't been able to interpret the Indus River valley civilization's writing system but have nevertheless acquired information about the civilization through historical artifacts. The fact that archaeologists have been able to learn about the Indus River valley civilization's customs and community organization from historical artifacts suggests that it isn't necessary to understand an ancient civilization's language to learn about the civilization.

Choice A is incorrect because the text doesn't discuss how easy it is to investigate ancient civilizations with or without knowledge of the civilization's language; rather, it states that even though researchers have not yet deciphered the language of the Indus River valley civilization, they are still able to learn about it through historical artifacts. Choice C is incorrect because the text doesn't make any claims as to what the focus of archaeological research should be. Rather, the text discusses how archaeologists have been able to learn about an ancient civilization through historical artifacts despite not understanding the civilization's language. Choice D is incorrect because the text states that the civilization's language has not yet been interpreted; it makes no mention of a debate about the language. Instead, the text suggests that examination of the historical artifacts has allowed archaeologists to learn about the civilization but has not aided thus far in deciphering its language.

Q6

A

A. is not conclusive evidence that the figure is Venus.

Choice A is the best answer because it presents the conclusion that most logically completes the text's discussion about the significance of the cupid found at Pompeii. The text indicates that the cupid is near a statue of a female figure who is fishing, and it goes on to indicate that because Venus is associated with cupids, some scholars believe the female figure to be the goddess Venus. But the text then says that, according to archaeologist Carla Brain, cupids may have also been associated with the activity of fishing, which, if true, would suggest that the mere appearance of a cupid near a female figure engaged in fishing does not indicate with certainty that the figure is Venus (that is, the cupid might be associated with fishing, and the figure might be anyone at all).

Choice B is incorrect because the text says nothing about how often Venus was depicted fishing in Roman art: it only implies that in certain instances a female figure may or may not be Venus. Choice C is incorrect because Carla Brain's proposed explanation for the presence of the cupids makes no reference to the female figure, and so the possibility that the figure in the artworks is in fact Venus cannot be definitively eliminated. Choice D is incorrect because there is nothing in the text to suggest that the only reasonable way to interpret the figure is as Venus.

Q7

A

A. makes civilian government jobs especially appealing to military veterans.

Choice A is the best answer because it presents the conclusion that most logically follows from the text's discussion of military veterans working in civilian government jobs in the United States. The text indicates that the proportion of military veterans working in civilian government jobs is considerably higher than the proportion of military veterans in the population as a whole. The text also notes that the unusually high representation of military veterans in these jobs may be a result of the organizational structures shared by civilian government entities and the military. Hence, it's reasonable to infer that it's the familiarity of the structures of civilian government that makes jobs there particularly attractive to military veterans.

Choice B is incorrect because the text doesn't address what a typical relationship between military service and later career preferences would be, and there's no indication that it's atypical for veterans to work in civilian government jobs after they've left the military. On the contrary, the text suggests that many military veterans are drawn to such jobs. Choice C is incorrect because the text is focused on the high representation of military veterans in civilian government jobs and doesn't address nonveterans or their possible interest in military service. Choice D is incorrect because the text conveys that military veterans may be particularly interested in civilian government jobs due to the familiarity of organizational structures that are already in place, but there's no reason to think that this interest would mean that more civilian government jobs will start to require military experience.

Q8

D

D. the timing of downbeats and offbeats may play a crucial role in giving a song swing.

Choice D is the best answer. The passage tells us that participants were “more than seven times likelier to characterize the modified songs as having swing than to characterize the original versions as having swing.” Because the modified songs had been changed by altering the timing of the downbeats and offbeats, this suggests that the timing of downbeats and offbeats may play a crucial role in giving a song swing.

Choice A is incorrect. Although the passage mentions that the researchers “synchronized the offbeats” in the modified songs, they also “delayed the downbeats” in those songs. Because we can’t disentangle whether it was the synchronized offbeats, the delayed downbeats, or the combination of both that increased the song’s swing, we don’t have enough information to make this inference. Choice B is incorrect. The passage doesn’t mention whether or not it’s difficult for a jazz musician to give a song swing, so there’s no basis for this inference. Choice C is incorrect. The passage doesn’t compare jazz songs that feature piano to those that don’t, so there’s no basis for this inference.

Q9

C

C. depictions of food in Dutch paintings of the period should be taken as realistic representations of Dutch eating habits.

Choice C is the best answer. The text tells us that lemons—an expensive imported product—feature in Dutch paintings of the period more frequently than dairy products do. Since it’s unlikely lemons were eaten more often than dairy, this fact casts doubt on the theory that these paintings realistically depicted Dutch diets.

Choice A is incorrect. In fact, the text suggests the opposite: it says that dairy products were a “major component of the Dutch diet,” while lemons were an expensive import item, so we can infer that lemons were eaten much more rarely than dairy products. Choice B is incorrect. The passage doesn’t mention painters from other countries, so there’s no basis to make this inference. Choice D is incorrect. In fact, the text suggests that Dutch painters do have reasons for depicting foods other than to show what Dutch people typically ate. For example, lemons in a painting could indicate wealth or prosperity.

Q10

B

- B. the timing of the initial formation of the middle ring played an important role in determining the eventual size of Earth.

Choice B is the best answer because it most logically follows from the text's discussion of André Izidoro, Rajdeep Dasgupta, and colleagues' computer simulations of our solar system's development. The text begins by stating that the simulations led the researchers to conclude that the solar system likely formed from three giant dust rings that encircled the Sun. The text explains that the four inner planets, including Earth, formed from the innermost ring and that the remaining planets formed from the middle ring. It then explains that in one simulation, the researchers delayed the formation of the middle ring—that is, they tested to see what would happen if the middle ring had formed later than it actually did. They found that doing so affected the size of the innermost planets, resulting in oversized super-Earths, planets that are much larger than Earth. Since the delayed timing had the effect of changing the size of Earth in the simulation relative to Earth's real size, it's reasonable to conclude that the timing of the middle ring's formation was important in determining Earth's eventual size.

Choice A is incorrect. Although the text explains that when the researchers delayed the formation of the middle ring in one simulation, the size of the innermost planets was affected (which suggests that the middle ring likely formed earlier than it did in this simulation), the text doesn't indicate that this was an initial simulation—that is, a simulation that was conducted before other simulations. Moreover, the text makes no reference to the specific results of any other simulations; therefore, there is no basis for comparing any conclusions based on the simulation in which the middle ring's formation was delayed with conclusions based on other simulations. Choice C is incorrect because the text discusses how altering the timing of the formation of the middle ring, not the outermost ring, affected the four innermost planets' eventual size in the researchers' simulation; therefore, the simulation offers no basis for a conclusion about how the outermost ring's formation affected the size of the planets. Choice D is incorrect because there is nothing in the text to suggest that the innermost ring produced all the solar system's planets. Rather, the text states that the simulations showed that the innermost planets formed from the innermost ring and that the remaining planets formed from the middle ring.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

